Ayush Srivastava

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Education

University of San Francisco — **San Francisco, CA**
M.S. in Data Science — *July 2025 - July 2026*

Birla Institute of Technology and Science, Pilani Campus — **Pilani, India**
M.Sc. Mathematics & B.E. Engineering — *Aug 2016 – May 2021*

Technical Skills

Tools / Packages: C/C++, Python, PyTorch, Hugging Face Transformers, QLoRA, BitsAndBytes, FAISS, Hugging Face, Cosine Similarity, AWS, Redshift, Matillion, SQL, MATLAB, Linux, Tableau, Scikit-Learn, Numpy, Pandas, Matplotlib, OS, NLTK, Dlib

Statistical Models: Linear / Logistic Regression, Decision Trees, Random Forests, K-means, Principal Component Analysis.

Experience

Data Scientist Intern — **San Francisco Bay Area, CA(Remote)**
LexisNexis USA — *Dec 2025 – Present*

- **Designed** and **deployed** an **API**-driven **LLM**-powered legal document summarization pipeline, incorporating human-in-the-loop evaluation, reducing manual review time by **60–70%** for downstream legal analysis workflows.

Senior Data Engineer — **Gurgaon, India**
Axtria-Ingenious Insights — *May 2023 – May 2025*

- **Led ETL development and feature engineering** across **6+ domains**, **productizing API wrappers** within an in-house ETL product to integrate **10+ reports** into Tableau, ensuring scalable, high-quality data architecture.
- **Processed and modeled 20M+ records** from **8+ vendors**, applying **statistical profiling and data mining techniques** to generate **30+ actionable insights** that informed strategy for **6+ pharmaceutical products**.
- Optimized data ingestion pipelines, boosting efficiency by **40%** via automation with **SQL**,**Python**,**Matillion**,**AWS Redshift**
- Collaborated with cross-functional teams to automate and integrate the delivery of **30+** Excel-based reports into **5+** dynamic, real-time dashboards using **Python** and **SQL**, ensuring seamless data flow and reporting automation.
- Performed data quality validation with **100+** test cases in **JIRA** to ensure defect-free pipelines.

Data Engineer — **Gurgaon, India**
Axtria-Ingenious Insights — *May 2022 – Apr 2023*

- **Spearheaded a team of 3 developers** to build Tableau dashboard visualizing **20+ KPIs** for **4+** teams improving sprint planning efficiency by **60%**; **Led 40+ client discussions** from a team of **50+** in 4 projects aligning business requirements
- **Enhanced BI data layers** by automating legacy reports, applying **data reconciliation**, improving accuracy by **15%**
- Managed weekly refresh of **100+** KPIs across **27+** dashboards; led **5+** weekly client requests with cross-functional teams.

Data Analyst — **Gurgaon, India**
Axtria-Ingenious Insights — *Jul 2021 – Apr 2022*

- Independently handled weekly refresh of **100+ Key Performance Indicators (KPIs)** on **27+** dashboards embedding statistical aggregation methods to ensure accuracy for **35+ sales representatives**.
- Led 5+ weekly ad-hoc requests as per client needs using **exploratory data analysis (EDA)**, partnering with commercial excellence, incentive compensation, onshore and data quality teams.

Data Scientist Intern — **Remote**
Sequretek IT Solutions Pvt. Ltd. — *May 2020 – Jul 2020*

- Developed **deep learning** models (VAE, LSTM) for **unsupervised time-series anomaly detection** across **150+** predictors and **1M+** Active Directory (AD) event logs, leveraging **statistical profiling** and **temporal trend analysis** in **Python** to identify cyberattacks, achieving **KL-Divergence loss** of **2.23/4.60** (train/validation).

Teaching Assistant — **New Delhi, India**
Coding Ninjas Pvt. Ltd. — *July 2019 – November 2019*

- Provided **personalized academic support** to students, addressing educational and conceptual gaps in **Data Structures and Algorithms (C++)**.
- Evaluated student performance through analysis of **test scores and assignments**, tracking progress and identifying improvement areas.
- Eliminated learning gaps and improved student outcomes using **structured problem-solving**, teaching aids, and targeted instructional strategies.

Publications and Projects

Legal Holding Generation using RAG Pipelines and Fine-Tuned LLMs — July 2025 - Aug 2025

- Developed a **RAG** pipeline on **LexGLUE** dataset using **Mistral 7B**, testing **3** retrieval strategies on **1,000** text input.
- Implemented **FAISS semantic retrieval** with **Legal-BERT**(**768 d**) embeddings, retrieving **3+** contexts per query in **RAG**.
- **Fine-tuned Mistral 7B** for generation on **LexGLUE** data (**1,000** records).**Cross-entropy loss** (Train: **1.46**, Test **1.63**).
- Quantized model to **4-bit** with **QLoRA**, training **33.5M** params, reducing size from **28GB** to **3.5GB** for efficient fine-tuning
- Fine-tuned LLM's cosine similarity: **0.8962**, RAG: Option 1:**0.8648**, Option 2:**0.8653**, Option 3:**0.8491**, on **200 test cases**.

Real Time Driver Drowsiness Detection Using GRU with CNN Features — Dec 2020

- Organized by: International Association of Pattern Recognition - Computer Vision and Image Processing 2020 — Springer, Singapore. DOI: https://doi.org/10.1007/978-981-16-1103-2_42
- Github: https://github.com/sayush2807/Driver-Drowsiness-Detection-using-GRU-with-CNN-features
- Built a **Custom Dlib Shape Predictor on iBUG-300W** to extract mouth region; **MAE** (Train**: 5.83,** Test: **11.83**).
- Trained 5 models: Inception-V3, DenseNet, Lenet, MobileNet, Feature Extractor on **AffectNet database (1M records)**.
- Developed Conv1D-GRU yawn detector achieving **99.99%** train and **99.97%** validation accuracy for real-time monitoring
- Achieved real-time performance of **30 FPS** on host and **23 FPS** on embedded board, ensuring in-vehicle system feasibility.
- Benchmarked **6+** facial datasets to select optimal sources for feature extraction, boosting model reliability and robustness.

Thesis | Image-Based Structural Damage Detection using CNN and Transfer Learning — Aug 2019 – Apr 2020

Supervisor: Prof. Ajit Pratap Singh

- Developed an **image classification pipeline** to detect **4 structural damage types** (no damage, flexural, shear, combined) using **CNNs and transfer learning** on **2,600+ images**.
- Implemented and compared **3 deep learning architectures**: custom CNN and **VGG16-based transfer learning models** with up to **6 dense layers and dropout**, improving model generalization.
- Applied **data augmentation** techniques (flip, rotate, zoom, skew) to address severe class imbalance, improving minority class representation from **8% to 26%** and eliminating classification bias.
- Achieved **91.47% test accuracy** and **99.64% training accuracy** using optimized VGG16 architecture, outperforming baseline CNN accuracy of **58.7%**.
- Trained models using **TensorFlow/Keras** with **Adam optimizer**, leveraging **GPU acceleration** (NVIDIA GTX 1650), and evaluated performance using **confusion matrix and classification metrics**.

Thesis | Traffic Flow Prediction using Deep Learning (LSTM, GRU, MLP) — Aug 2020 – May 2021

Supervisor: Prof. Ajit Pratap Singh

- Developed **deep learning models** for **time series traffic flow prediction** using **12,000+ sequential data points** from California PeMS dataset with 5-minute aggregation intervals.
- Implemented and compared **Multi-Layer Perceptron (MLP)**, **GRU**, and **LSTM** architectures using **TensorFlow/Keras**, leveraging **GPU acceleration** (NVIDIA GTX 1650).
- Designed supervised time series pipeline using **sliding window technique (window size=8)** to capture temporal dependencies and improve sequence prediction accuracy.
- Achieved best performance using **LSTM model with RMSE: 9.81 and R^2: 0.9398**, outperforming MLP (RMSE: 11.03) and GRU (RMSE: 10.08), demonstrating superior temporal modeling capability.
- Built deep recurrent architecture with **3 LSTM layers and dense layers**, optimizing hyperparameters using **Adam optimizer**, improving prediction accuracy by **11% vs baseline MLP**.

Comparative Analysis of Neural Networks and ANFIS in Water Storage Prediction — Jan 2020 – May 2020

- Collected weekly time series data of live storage of Mettur Dam, Chennai (2001-2013) to train 2 forecasting models.
- **Long Short-Term Memory (LSTM) model** using **Keras** in **Python**: Training: 2.44, Validation: 2.66, Testing: 7.87.
- **Adaptive Neuro-Fuzzy Inference System (ANFIS)** in MATLAB: Training: 4.42, Validation: 5.09, Testing: 8.12.

Face Generation, Udacity — September 2020 – October 2020

- Trained a **Deep Convolutional Generative Adversarial Network (DCGAN)** on the **CelebFaces Attributes Dataset (CelebA)** to generate realistic human faces.
- Implemented **Discriminator** and **Generator** architectures using **Convolution** and **Transpose Convolution** layers respectively in **PyTorch**.
- Achieved **Discriminator loss: 0.06** and **Generator loss: 0.56**, producing high-quality synthetic face images using the trained Generator.

Dog Breed Classification using CNNs and Transfer Learning (PyTorch) — Sep 2020 – Oct 2020

- Built an end-to-end **computer vision pipeline** using **Convolutional Neural Networks (CNNs)** and **transfer learning** in **PyTorch** to classify dog breeds from real-world images.
- Developed an intelligent image classification system capable of **detecting dog breeds and identifying resembling dog breeds for human faces**, demonstrating multi-class classification capability.

- Implemented deep learning models leveraging **pretrained architectures** and fine-tuned fully connected layers, improving classification performance and convergence speed.
- Integrated **Haar Cascade classifiers** for human face detection and combined with CNN-based dog breed classification to build a complete inference pipeline.
- Designed modular prediction pipeline deployable in **web/mobile applications**, enabling real-time inference on user-supplied images.

TV Script Generation using LSTM-based Recurrent Neural Networks **Aug 2020 – Sep 2020**

- Developed a **text generation model** using **Recurrent Neural Networks (RNNs)** and **Long Short-Term Memory (LSTM)** architecture to generate realistic TV scripts from sequential text data.
- Trained deep learning models on the **Seinfeld TV scripts dataset (9 seasons)**, enabling the model to learn linguistic structure, dialogue patterns, and contextual dependencies.
- Implemented complete **NLP preprocessing pipeline**, including tokenization, vocabulary encoding, sequence batching, and supervised sequence prediction.
- Built and trained multi-layer **LSTM network** in **Python**, generating coherent multi-character dialogues and long-form script content.
- Designed inference pipeline to generate new scripts from learned probability distributions, demonstrating practical applications in **generative AI and language modeling**.

Certifications

Deep Learning Specialization, Deeplearning.ai, Coursera **Apr 2020 – May 2020**

Machine Learning & DSA in C++, Coding Ninjas Pvt Ltd **May 2019 – Aug 2019**

TensorFlow in Practice Specialization, DeepLearning.AI (Coursera) **May 2020**

Position of Responsibility

Sports Secretary, Malviya Bhawan — Sports Council **BITS Pilani, India**
August 2017 – May 2018

- Developed financial records for **10+ sports events**, ensuring **100% accuracy** and compliance with budgetary guidelines.
- Managed an annual budget of **2 lacs** to procure sports kits and equipment for **250+ hostel residents**, increasing participation by **30%**.
- Coordinated logistics and planned events for the **BITS Open Sports Meet**, attended by **500+ participants** from **10+ colleges**.
- Organized intra-hostel tournaments across **3 sports**, with participation from **80+ students** across **6 hostels**.

Achievements

1st Place, Cricket Tournament — **BITS Open Sports Meet (BOSM)**, BITS Pilani, 2018.

1st Place (2×), Badminton Tournament — **Intra-BITS Open Sports Meet (I-BOSM)**, 2017 and 2019.

Selected to represent **Mount Carmel School** at the **Kennedy Space Center, NASA**, Florida, USA, 2013.

Gold Medalist, **National Abacus Championship Tournament**, BrainoBrain, 2010.